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Vibration insensitive optical cavity

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An optical cavity is designed and implemented that is insensitive to vibration in all directions. The cavity is mounted with its optical axis in the horizontal plane. A minimum response of 0.1 (3.7) $\mathrm{kHz/ms^{-2}}$ is achieved for low-frequency vertical (horizontal) vibrations.

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I. INTRODUCTION

Ultrastable lasers are a key element in optical frequency standards where they are used both to probe an atomic transition and to provide a measurable output. A laser oscillator is stabilized to a passive optical cavity and, given a sufficiently high fidelity of control, the frequency is defined by the cavity. Thus, the dimensional stability of the cavity becomes paramount. Accelerations acting upon the cavity cause it to distort and typically the sensitivity is $\sim 100\ \mathrm{kHz/ms^{-2}}$. With quiet conditions and vibration isolation, subhertz laser linewidths can be achieved [1–4]. However, accelerations remain the dominant source of instability on the 0.1–10 s time scale. If one is ever to envisage such an apparatus leaving the confines of a specialized laboratory, the extreme sensitivity to vibrations must be overcome. Also, the availability of ultranarrow atomic references [5], where the observed Q is determined by the laser linewidth, motivates the pursuit of even greater stability. One could try to isolate the cavity further from vibrations, however, given the range of contributing frequencies (<10 Hz), the difficulty and expense will far outweigh the return in performance. A better approach by far is to reduce the *sensitivity* of the cavity to vibrations, and indeed this route has already met with some success.

An optical cavity typically comprises two highly reflective concave mirrors bonded to a spacer with a central bore and has axial symmetry. The optical mode coincides with the symmetry axis and has a small diameter at the mirror surface relative to the cavity dimensions. To achieve vibration insensitivity, one has to ensure that the distance between the two points at the centers of the mirror surfaces remains invariant when a force is applied to the support. This may simply be achieved through symmetrical mounting: a force applied through the mount will cause one half of the cavity to contract while the other half expands with the result that there is no net change in dimension. An obvious implementation of this is to hold the cavity about its plane of symmetry with its axis aligned with gravity and, using this approach, a reduced acceleration sensitivity of $10\ \mathrm{kHz/ms^{-2}}$ has been demonstrated [6]. A cavity mounted with its axis horizontal sags asymmetrically under its own weight. However, through optimization of the support positions, the distance between the centers of the mirrors can be made invariant, even though the cavity still deforms on application of a vertical force [7] and, by this method, a vertical acceleration sensitivity of $1.5\ \mathrm{kHz/ms^{-2}}$ has been demonstrated [8]. A similar result is achieved by removing material from the underside of the cavity and using this approach we report a vertical acceleration sensitivity $<0.1\ \mathrm{kHz/ms^{-2}}$.

II. DESIGN

The geometry is shown in Fig. 1. Square "cutouts" are made to the underside of a cylindrical spacer and the cavity is supported at four points. The cutouts compensate for vertical forces. Vibration insensitivity in the horizontal plane is

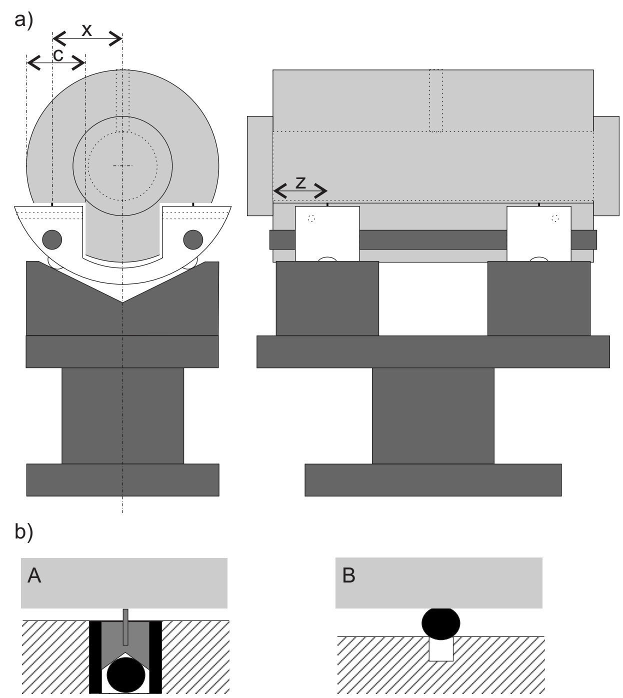

FIG. 1. (a) Cutout cavity on mount. The cavity is represented by the light gray shaded area. The support "yokes" are shown in white. Spacer length, 99.8 mm; diameter, 60 mm; axial bore diameter, 21.5 mm; vent-hole diameter 4 mm. The mirrors are 31 mm in diameter and 8 mm thick. The parameters relevant to the design, the cut depth c and the support coordinates x and z are indicated. (b) Details of the support point in cross section. The black shaded areas represent rubber tubes and spheres. Case A: Diamond stylus set into a cylindrical ceramic mount, cushioned on all sides by a rubber tube, and from below by a rubber sphere. The stylus digs into the underside of the cavity, and can be considered to be in rigid contact with it. Case B: 3-mm-diam rubber sphere recessed into a cylindrical hole in the yoke.

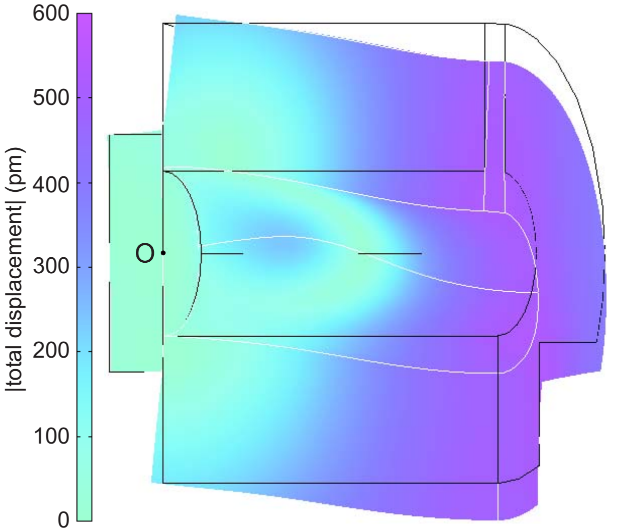

FIG. 2. (Color online) Plot showing the deformation of a quarter section of the cavity. The point O marks the center of the mirror surface. $c=18.45$ mm, $x=22$ mm, and $z=17$ mm. The scale of the distortion is amplified by a factor of 10^7 and a uniform downward displacement of 10 nm is subtracted. The white (black) lines indicate the outline of the loaded (unloaded) geometry.

achieved through symmetrical mounting. A finite-element analysis has been carried out for this design using commercially available software [9]. A static stress-strain model is used, because the frequency of accelerations that make a significant contribution to the frequency noise are less than 10 Hz and therefore, to a good approximation, can be considered to be at dc relative to the first structural resonance, which is $\sim$10 kHz. The model considers just a quarter section of the cavity (excluding the mount), exploiting the symmetry of the geometry in order to reduce computation time. The planes of symmetry are restrained not to move normal to their surfaces, thus only symmetric solutions are considered. The support is modeled as a point on the horizontal cutout surface. A body force due to gravity is applied throughout and the support is constrained not to move in the vertical direction. This is entirely equivalent to a model where a vertical force is applied through the support, but is simpler to implement. A pseudorandom tetragonal mesh is generated comprising approximately 6000 mesh elements with 30 000 degrees of freedom. The mirror substrates and spacer are made from ultra-low-expansivity glass and values of 67.6×10^9 Pa and 0.17 are used for Young's modulus and Poisson's ratio, respectively.

The solution for total displacement is shown in Fig. 2. Although the cavity deforms when loaded, the axial displacement at the center of the mirror surface, w_O, remains unchanged. Figure 3 shows w_O as a function of the support position and cut depth. One can see that it does indeed pass through zero for certain combinations of x, z, and c. There is in fact no unique null condition: the locus of the displacement zero is a plane in xzc space. The existence of zero crossings in the displacement as a function of support position is an important aspect of the design: it offers an experimental parameter that can be adjusted to obtain the null condition. A design was chosen with $c=18.45$ mm and $x=22$ mm, with the parameter z left free to be varied. The solution for these parameters is represented by the open circles in Fig. 3. For a detailed description of finite-element-based analysis of cavity deformation under the influence of vibration noise, for a number of different cavity and support configurations, one is also referred to the work of Chen *et al.* [10].

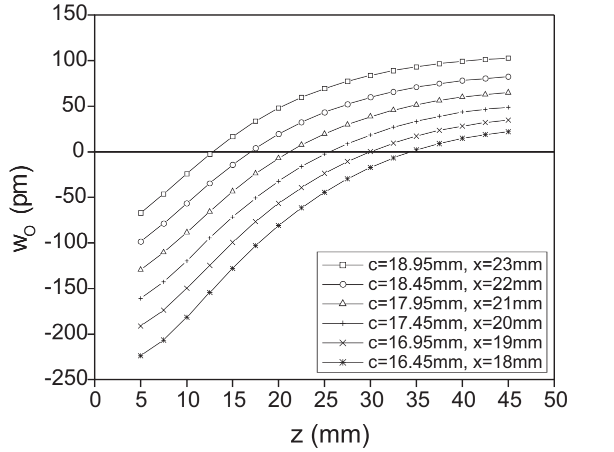

FIG. 3. Plot of axial displacement at the center of the mirror surface, w_O, as a function of z for various c and x.

III. IMPLEMENTATION

A cavity and mount have been made to the design shown in Fig. 1. The mirrors and spacer are all made from ultra-low-expansivity glass. The mirrors have a concave surface with a radius of curvature of 350 mm and have a reflectivity of 0.999 98. These are optically contacted to the spacer, concentric with its axis. The cavity is placed symmetrically on the mount and the whole cavity-support structure has two-fold horizontal symmetry. The yokes (see the caption of Fig. 1) fix x and small grub screws set within them can nudge the cavity from either side, ensuring symmetrical placement with respect to the mount. Symmetrical placement along the axis is done by eye and is accurate to within half a millimeter. The yokes can slide along the parallel rods, which ensure that symmetry is maintained while allowing adjustment of z. To make this adjustment, the cavity has to be lifted off and then replaced back on the support.

The difficulty with having four contact points is ensuring symmetry of the restoring forces acting at these points. With this in mind, the support is constructed so that forces acting on the base are split equally between the contact points. Figure 1(b) shows the two types of contact used in this work. In both cases the contact is made compliant by the use of rubber tubes and/or spheres. As these are by far the most elastic elements of the whole support, they in fact define the compliance. This simplifies the task of achieving force symmetry: all one has to ensure is that the stiff part of the mount is symmetric and that the compliant elements are identical. High compliance also has the advantage that the extent of the compression under loading is large compared to the dimensions of any residual asymmetries, thus their effect is diminished.

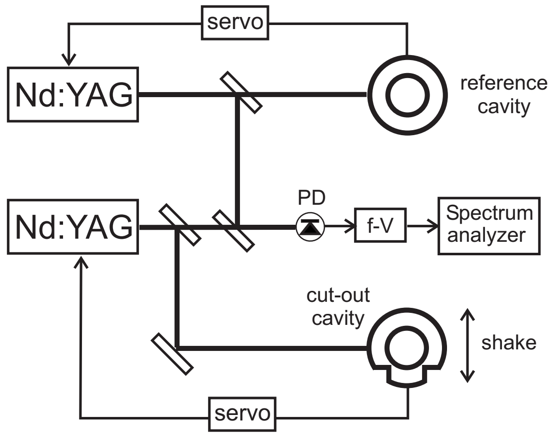

FIG. 4. Schematic of experimental setup for measuring vibration response. PD: photodiode; f-V: frequency-to-voltage converter. The thicker lines indicate beam paths; the thinner lines indicate electronic signal paths.

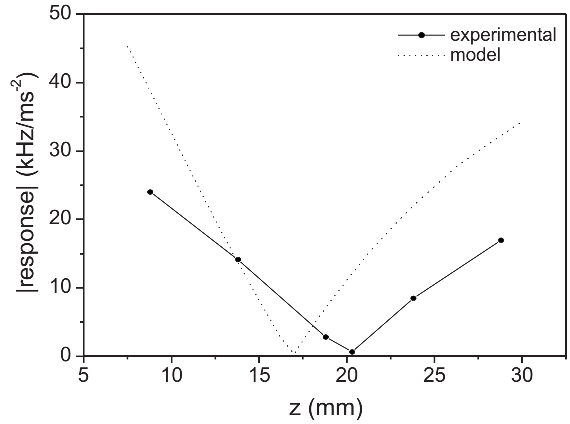

FIG. 5. Plot of experimental response of cutout cavity to vertical acceleration as a function of the support position. $c=18.45$ mm, $x=22$ mm, and $z=17$ mm. The response predicted by the model is also shown. This is calculated by converting the displacement to the frequency change this would induce, taking into account that the center points of the two mirrors are displaced by equal and opposite amounts.

IV. EXPERIMENT

The response of the cavity to vibrations is measured using the setup shown in Fig. 4. Two Nd:YAG lasers at 1064 nm are each frequency controlled to an optical cavity. Light from both lasers is combined on a fast photodiode and a beat note obtained, thus the relative frequency fluctuations between the two systems can be measured. The power spectrum of these fluctuations is analyzed following frequency-to-voltage conversion. Both cavities are placed on active vibration isolation (AVI) platforms. In the case of the reference cavity, the isolation is engaged, whereas for the cutout cavity, the AVI is used in reverse as a "shaker": a sinusoidal acceleration can be applied along any one of three orthogonal axes to which the axes of the cavity are aligned. The amplitude of the acceleration on the platform is measured using a three-axis seismometer. Some mixing between the degrees of freedom occurs at both low and high frequencies, therefore, a drive frequency of 15 Hz is used. This gives a relatively pure acceleration in the direction of choice, while being sufficiently low for the response to be considered quasistatic. Frequency modulation at the drive frequency can be resolved in the power spectrum, and from the modulation amplitude and measured acceleration, a response is derived.

V. RESULTS AND DISCUSSION

Figure 5 shows the response to vertical vibrations as a function of support position. The minimum value is 0.6 kHz/ms^{-2} for $z=20.3$ mm. The response to horizontal vibrations is more difficult to null, depending critically upon the symmetry of the mount, not only in its geometry, but particularly in the restoring force at each point. With the cavity mounted on diamond styli, (Fig. 1, case A), the null in the vertical direction is good (as shown in Fig. 5), however, the horizontal response varies in the range 10–100 kHz/ms^{-2}. On changing to rubber spheres (Fig. 1, case B), the horizontal response is 3.7 (0.9) kHz/ms^{-2} in line with (perpendicular to) the cavity axis, while the vertical response is <0.1 kHz/ms^{-2}. A better result is achieved with the spheres because they are more horizontally compliant than the rubber tubes and also they are closer to being geometrically identical than the combination of components used in case A. It is likely that in case A there is some slack in the fit of the ceramic within the rubber tube and the recess in the yoke, and this asymmetry, coupled with greater horizontal stiffness, leads to poor reproducibility.

A potential problem with using elastic contact points is that the whole support becomes resonant. This has been tested by driving the AVI at higher frequencies and the response is indeed found to be broadly resonant, amplified by up to a factor of seven at 40 Hz. This however, is not as serious a problem as it may seem. At frequencies above 10 Hz, the AVI has an attenuation of 35 dB, thus the increased response is more than compensated for by the high degree of isolation. Accelerations at frequencies below 10 Hz make the dominant contribution to the frequency noise, however, the attenuation of the AVI falls off in this range and it ceases to isolate below ~2 Hz. For the vibration insensitivity to be of use, it is vital that the effective dc response of the cavity be reduced and this is accomplished by the cutout design.

Comparing the experimental data in Fig. 5 with the model, one can see that there is good qualitative agreement, while some quantitative discrepancy remains. The model predicts a null at lower z and the gradient about the null is also larger. The dependence of the null on experimentally adjustable parameters diminishes the importance of the model's accuracy, as a null can be found over a wide range of values. However, for future work it would be useful to discover the origin of the discrepancy. The model is relatively simple: it assumes symmetry and does not include mirror curvature or dependence on tilt and this may account for the error.

VI. CONCLUSION

A reduction in the response of an optical cavity to vertical vibrations by more than a factor of 1000 has been demonstrated experimentally, aided by finite-element design. The technique relies upon compensation of the deformation of the cavity structure such that the length of the optical axis remains constant under applied acceleration. The horizontal response has been reduced by around a factor of 30, and this relies upon the mounting being symmetrical, not only in its geometry, but also in the restoring forces acting at the contact points. It has been shown that this is best achieved by mounting the cavity on rubber spheres.

The results presented here offer more than two orders of magnitude lower sensitivity to vertical vibrations compared to the work of Notcutt *et al.* [6] and more than an order of magnitude (more than a factor of three) lower sensitivity to vertical (horizontal) vibrations compared to the work of Nazarova *et al.* [8]. In terms of whether it is best to mount the cavity with its axis vertical or horizontal, the cavity is most sensitive to vibrations along its axis, therefore, one might consider it best to align the axis with the direction of least vibration. For the AVI platforms used in this work there is little dependence of the magnitude of acceleration on direction, therefore, in this respect, there is no preferred orientation.

With the effect of vibrations suppressed, the frequency stability of light controlled to the cavity will become limited by dimensional changes due to thermal fluctuations of the mirror substrates and their coatings [11]. With regard to suppressing these fluctuations, it would be better for the cavity to be longer, such that they become a smaller fraction of the cavity's total optical length. Nulling the vibration response relies upon accurate positioning of the support points and balancing of the restoring forces at those points. Since the absolute experimental errors in position or force are unlikely to change significantly upon increasing the cavity's length, and the cavity's strain is proportional to each of these absolute errors, then the fractional frequency change will be independent of length. The cavity length may thus be increased while maintaining the same vibration insensitivity with the advantage that the effect of thermal fluctuations will be reduced.

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